

Organization

Dr. Christian A. Mahringer

TUM School of Management, Campus Heilbronn



Course agenda

Date	Main theme
13.04.2026	Definitional basics of organization
20.04.2026	Formal organizational design
27.04.2026	Organization and environment
04.05.2026	Strategic organization design
11.05.2026	Technology and organization
18.05.2026	Information technology, AI, and organization
01.06.2026	Informal organization

Date	Main theme
08.06.2026	Informal organization
15.06.2026	Dynamic perspectives on organizing
22.06.2026	Guest lecture, Felix Ehrenfels, mmmake
06.07.2026	Trends in organization design
13.07.2026	Trends in organization design
17.07.2026	Review and wrap-up

Session 06

AI and organization:

How a new technology shapes everyday work

What you should be able to do after this session

Learning goals

- Define AI in organizations.
- Explain why AI effects are not only technologically determined.
- Analyze how AI shapes tasks, roles, expertise, routines, and ethics.
- Apply insights about AI to a practical case.



Image source: ChatGPT

Opening activation

When an organization introduces AI, what changes in the organization?



Image source: ChatGPT

Session roadmap



Today's structure

1) What AI means in organizations

2) Effects of AI on organizations

3) Case study

Session roadmap

Today's structure

A thick green vertical bar is positioned to the left of the first item.

1) What AI means in organizations

A thick blue vertical bar is positioned to the left of the second item.

2) Effects of AI on organizations

A thick blue vertical bar is positioned to the left of the third item.

3) Case study

Working definition

AI in organizations



“Artificial intelligence (AI) refers to machines performing cognitive functions that are usually associated with human minds, such as learning, interacting, and problem solving”

AI is neither a stable object nor a new topic

- Earlier expert systems followed explicit rules.
- Contemporary AI often learns from data and adapts behavior.
- Different AI systems rely on different logics.
- Different types of AI can produce different organizational effects.

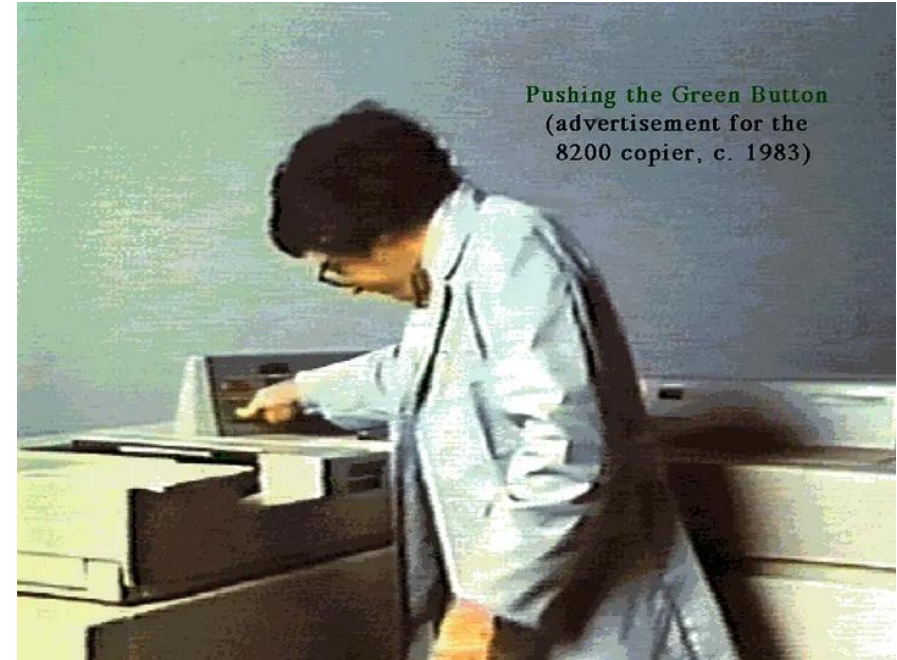


Image source: <https://medium.com/@emmet.obriain/the-big-green-button-myth-resisting-dehumanised-research-the-corporate-constraint-of-a0da92de48a9>; See: Suchman, L. A. (1985). Plans and situated actions: The problem of human-machine communication. Xerox Corporation.

Session roadmap

Today's structure



1) What AI means in organizations



2) Effects of AI on organizations



3) Case study

No determinism

A core premise for studying AI in organizations

AI does not produce one fixed organizational effect. Effects depend on:

- how AI is developed
- how AI is introduced
- how AI becomes embedded in everyday work
- how people interpret and use outputs
- how routines, roles, and responsibilities are adjusted
- how social relations and power relations are reorganized

→ **AI should be studied in organizational practice, not as an isolated technical object.**

Huysman, M. (2026). Studying AI in the wild: Reflections from the AI @work research group. *Journal of Management Studies*, 63(2), 315–334.

Four effect domains

Where AI becomes organizationally consequential

Task division

Roles, knowledge and expertise

Organizational stability and change

Fairness, ethics, and accountability

Four effect domains

Where AI becomes organizationally consequential



Task division



Roles, knowledge and expertise



Organizational stability and change



Fairness, ethics, and accountability

Automation and augmentation

Two basic modes of AI use

Automation

- Machines take over a human task.
- Human action is partly replaced.



Augmentation

- Humans and machines work together.
- Human action is supported or extended.



Raisch, S., & Krakowski, S. (2021). Artificial intelligence and management: The automation–augmentation paradox. *Academy of Management Review*, 46(1), 192–210.

The automation-augmentation paradox

Automation and augmentation are interdependent rather than simple opposites

Augmentation prepares automation

Humans train, improve, and validate systems.

Automation prompts augmentation

Adjacent tasks may require human judgment.

Both can coexist at the same time

Different routine parts may differ.

One sided focus is risky

Automation can weaken expertise.
Augmentation can bind resources.

Raisch, S., & Krakowski, S. (2021). Artificial intelligence and management: The automation–augmentation paradox. *Academy of Management Review*, 46(1), 192–210;
Mahringer, C. A., Danner-Schröder, A., Müller-Seitz, G., & Renzl, B. (2024). How does artificial intelligence promote change and stability of organizational routines? The role of automation and augmentation. *Journal of Competences, Strategy & Management*, 12, 1–17.

Automation and augmentation in hiring

Task:

- Imagine AI is introduced to support personnel hiring in a medium-sized industry firm.
- Analyze which hiring task is automated, which task is augmented, which adjacent task changes, and the setup creates task splitting or task integration.
- Identify three key challenges in redesigning the task.

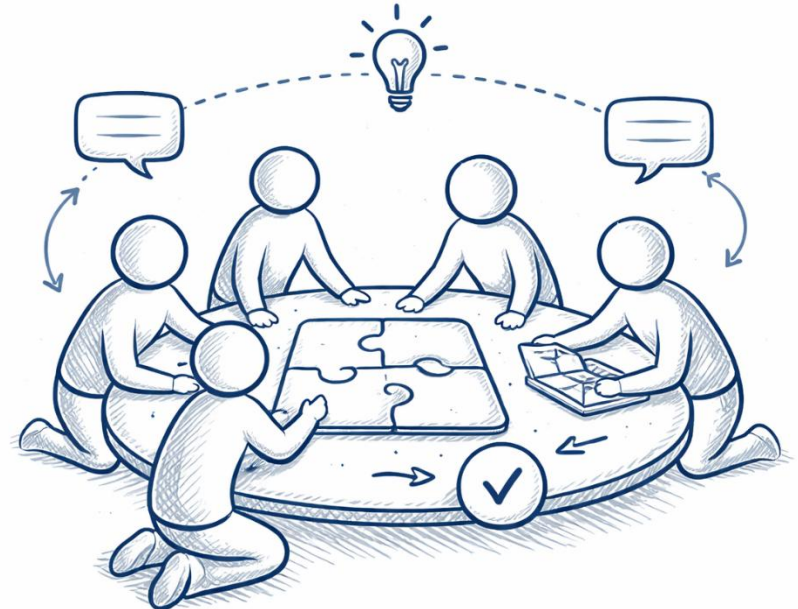
Goal:

- Deepen your understanding of how tasks are automated and augmented.

Modus:

- Pair work.

Image source: ChatGPT



Four effect domains

Where AI becomes organizationally consequential

Task division

Roles, knowledge and expertise

Organizational stability and change

Fairness, ethics, and accountability

New roles around AI

How humans assist machines

Training

People train AI systems through data, examples, criteria, and interaction design.

Explaining

People explain AI outputs, especially when results are opaque or counterintuitive.

Sustaining

People ensure responsible use, including safeguards, privacy, and ethical control.

Wilson, H. J., & Daugherty, P. R. (2018). Collaborative intelligence: Humans and AI are joining forces. *Harvard Business Review*, 96(4), 114–123.

New roles around AI

How machines assist humans

Amplifying

AI strengthens analytical and creative capabilities by providing information, options, or proposals.

Interacting

AI enables new interactions with customers, users, or employees.

Embodying

AI extends physical capabilities through robotics and embodied systems.

Wilson, H. J., & Daugherty, P. R. (2018). Collaborative intelligence: Humans and AI are joining forces. *Harvard Business Review*, 96(4), 114–123.

New roles around AI

Centaurs versus cyborgs

Centaurs

- Human and AI work on different parts of the task.
- Human decides what to delegate to AI.
- **Mode:** division of labor.
- **Risk:** underusing AI's support.



Cyborgs

- Human and AI co-produce throughout the task.
- Human continuously prompts, edits, and steers AI.
- **Mode:** deep integration.
- **Risk:** over-relying on AI output.



Dell'Acqua, F. et al. (2023). Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. SSRN Electronic Journal, 1–58; Image source: ChatGPT

Expertise effects

AI can grow or erode expertise

Expertise growth

- AI provides information, variants, forecasts, and analytical support.
- AI can require new digital, data analytical, decision-related, and learning skills.

Expertise erosion

- AI can remove learning opportunities.
- AI can separate production from interpretation.
- AI can create dependence on poorly understood outputs (cognitive shortcuts).

Anthony, C. (2021). When knowledge work and analytical technologies collide: The practices and consequences of black boxing algorithmic technologies. *Administrative Science Quarterly*, 66(4), 1173–1212.

Example: AI for automated financial data analysis

AI can grow or erode expertise

	Partitioners	Co-Constructors
Seniors' beliefs about AI	AI supports efficiency.	AI is transformative and requires caution.
Task allocation	Task splitting: Junior bankers produce analyses. Seniors interpret them.	Task integration: Junior bankers produce analyses. Both juniors and seniors are responsible for interpretation.
Seniors' actions	• Emphasize strict adherence to procedures. • Check internal consistency. • Dictate changes.	• Ask for defensible analysis. • Scrutinize the analytical process.
Consequences	• Knowledge outputs depend on AI tools. • Junior bankers do not develop expertise.	• Knowledge outputs are consistent. • Junior bankers develop expertise.



Anthony, C. (2021). When knowledge work and analytical technologies collide: The practices and consequences of black boxing algorithmic technologies. *Administrative Science Quarterly*, 66(4), 1173–1212.

The opportunities and risks of AI at university

Students often use AI to complete course tasks and prepare for exams.
Where do you think are the risks and opportunitiess of this usage?



Image source: ChatGPT

Four effect domains

Where AI becomes organizationally consequential

Task division

Roles, knowledge and expertise

Organizational stability and change

Fairness, ethics, and accountability

How AI contributes to stability

AI-based routines can reinforce inertia

Mechanism	Explanation	Empirical example
Bounded retheorization	The organization adjusts the model incrementally, but does not rethink its underlying assumptions.	Moody's modified its M3 model for subprime mortgages, but largely retained existing assumptions about housing prices, borrower behavior, and default risk.
Sedimentation of assumptions	Old assumptions become embedded in data inputs, even when the environment has changed.	Moody's continued to rely on FICO scores, loan tapes, and historical mortgage data, although documentation quality and lending standards had deteriorated.
Simulation of the unknown future	Algorithmic simulations create confidence that future risks are covered, but they may miss unprecedented shifts.	Moody's simulated many macroeconomic scenarios, yet the model did not capture the possibility of a sharp nationwide housing price collapse.
Specialized compartmentalization	Different actors handle separate parts of the model, weakening holistic understanding and accountability.	Model developers, analysts, rating committees, and surveillance teams worked on different parts of the rating routine, making it harder to question the model as a whole.

Omidvar, O., Safavi, M., & Glaser, V. L. (2023). Algorithmic routines and dynamic inertia: How organizations avoid adapting to changes in the environment. *Journal of Management Studies*, 60(2), 313–345.

How AI contributes to stability and change

Mechanism			Examples from business practice
Change	Capacitating	Artificial intelligence enables new actions and relations that were not possible before or that required excessive resources.	'Merck' uses AI to speed up or even enable the innovation process when it comes to drug development. Towards this end, the predictive models not only forecast the viability of a novel compound, but it becomes possible to obtain an explanation why a certain compound will work or why not. Hence, the AI performs new actions in the product development routine, which were difficult to perform without AI.
	Reframing	Artificial intelligence helps actors to revise their previously held conceptions about the pattern of actions of a routine, which helps them see it in a new way.	'Celonis' provides an AI that can automatically detect patterns in business processes from vast amounts of data and suggest how to change those processes. Hence, it may suggest new patterns of actions that have not been envisioned before.
Stability	Shielding	Artificial intelligence blocks influences that could change actions and relations.	'Intel' uses a highly automated production facility for their chip production. Here, it may be difficult for staff to observe and reflect on the actions carried out by the robots. The robots may not change their actions.
	Adhering	Artificial intelligence prevents actors from deviating from established actions, due to sequential interdependence or escalating commitment.	In the 'BMW i3' assembly in Leipzig, a robot works directly with the employees on the assembly line. The robot is applying the adhesive to the windshield before the human employee overtakes. Due to the close interaction, the human actor needs to adhere to the action pattern.

Mahringer, C. A., Danner-Schröder, A., Müller-Seitz, G., & Renzl, B. (2024). How does artificial intelligence promote change and stability of organizational routines? The role of automation and augmentation. *Journal of Competences, Strategy & Management*, 12, 1–17.

Stability and change in practice

In your experience, did you observe tasks that changed through AI or that should have changed, but didn't?



Image source: ChatGPT

Four effect domains

Where AI becomes organizationally consequential

Task division

Roles, knowledge and expertise

Organizational stability and change

Fairness, ethics, and accountability

Risks of AI in knowledge work

Multipurpose usability

Employees can use GenAI for many tasks and may bypass experts.

Unpredictable plausibility

Outputs can sound convincing while being wrong or incomplete.

Hyper personalizability

AI assistants can become tailored to individuals and reduce shared interaction.

Example: AI-generated consulting reports

Search

FORTUNE

Subscribe for \$1Sign in

HomeLatestFortune 500FinanceTechLeadershipLifestyleRankingsMultimedia

AI • DELOITTE

Deloitte allegedly cited AI-generated research in a million-dollar report for a Canadian provincial government

By Nino Paoli

Former News Fellow

November 25, 2025, 6:03 AM ET

Add us on



London-based consulting firm Deloitte has come under fire for the second time this year after another foreign government-commissioned report contains potentially AI-generated errors.

J. DAVID AKE/GETTY IMAGES

<https://fortune.com/2025/11/25/deloitte-caught-fabricated-ai-generated-research-million-dollar-report-canada-government/>

Dr. Christian A. Mahringer | TUM School of Management, Campus Heilbronn

33

Algorithmic control

AI can move from support to control

Algorithmic management can include:

- monitoring work behavior
- comparing behavior with expected patterns
- allocating tasks
- evaluating performance
- triggering rewards, sanctions, or restrictions

→ **These changes reduce worker control and discretion.**

Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals*, 14(1), 366–410.

Fairness with AI

Fairness is enacted in practice

Fairness emerges through how people:

- define values
- embed them in AI procedures
- perform them in practice
- defend them through professional mandates
- stabilize them through scores, thresholds, and metrics

Ethical risks

Core organizational risk areas

Bias

AI can reduce human bias, but also scale machine bias.

Privacy

AI relies on large amounts of data and raises questions about personal data control.

Transparency

Opaque decisions can undermine trust and make explanation difficult.

Accountability

Responsibility becomes unclear when decisions depend on data, models, and automated outputs.

→ **Machines should challenge humans, and humans should challenge machines.**

Benbya, H., Davenport, T. H., & Pachidi, S. (2020). Artificial intelligence in organizations: Current state and future opportunities. MIS Quarterly Executive, 19(4), 9–21.

AI use offers paths to realizing risks and opportunities



Image source: ChatGPT

Analytical framework: How AI becomes consequential

Effect domain	Diagnostic question	Key concepts to apply	What students should look for in the case
Task division	Which parts of work are automated, augmented, split, or integrated? What are the benefits and risks?	Automation, augmentation, automation–augmentation paradox, centaur vs. cyborg work	Tasks taken over by AI, tasks supported by AI, adjacent tasks that change, new dependencies between humans and AI
Roles, knowledge, and expertise	How does AI redistribute expertise and change what people need to know? Which roles become obsolete and which roles emerge?	Training, explaining, sustaining, amplifying, interacting, embodying, expertise growth, expertise erosion	New human roles around AI, loss of learning opportunities, new analytical skills, dependence on AI outputs, meaningful or superficial human involvement
Organizational stability and change	Does AI enable routine change, or does it reinforce existing patterns? Which mechanisms drive these dynamics?	Capacitating, reframing, shielding, adhering, bounded retheorization, sedimented assumptions, simulation of unknown future, specialized compartmentalization	New actions made possible by AI, old assumptions embedded in data or models, routines that become harder to question, changes that stall
Fairness, ethics, and accountability	Who becomes responsible for AI-based decisions, and which risks emerge?	Bias, privacy, transparency, accountability, algorithmic control, enacted fairness	Unclear responsibility, opaque decisions, biased or scaled decisions, reduced discretion, values embedded in thresholds, scores, and procedures

Session roadmap

Today's structure

A solid blue vertical bar is positioned to the left of the first item.

1) What AI means in organizations

A solid blue vertical bar is positioned to the left of the second item.

2) Effects of AI on organizations

A solid green vertical bar is positioned to the left of the third item.

3) Case study

Case Study

Task:

- Solve the case study in Moodle. Each team focuses on one effect domain.
- Use the concepts from the analytical framework explicitly in your diagnosis.
- Create a [presentation](#) with your results.

Goal:

- Deepen your understanding AI effects on organizations.

Modus:

- Group work.

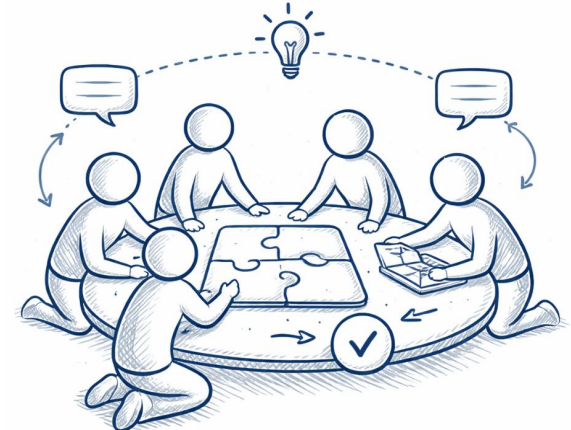


Image source: ChatGPT

Feedback

Please fill out the survey



Takeaways from today

AI is a specific technology, which mimics human intellectual capabilities.

Similar to other technologies, AI's effects depend on how it is brought into use.

We discussed four central effect domains: (1) Task division, (2) roles, knowledge and expertise, (3) stability and change, (4) fairness, ethics, and accountability.

You saw how applying concepts related to these effect domains can help you solve practical problems.

If technologies' effects depend on use, we need to examine the informal organization that provides the context for such use.

Course agenda

Date	Main theme
13.04.2026	Definitional basics of organization
20.04.2026	Formal organizational design
27.04.2026	Organization and environment
04.05.2026	Strategic organization design
11.05.2026	Technology and organization
18.05.2026	Information technology, AI, and organization
01.06.2026	Informal organization

Date	Main theme
08.06.2026	Informal organization
15.06.2026	Dynamic perspectives on organizing
22.06.2026	Guest lecture, Felix Ehrenfels, mmmake
06.07.2026	Trends in organization design
13.07.2026	Trends in organization design
17.07.2026	Review and wrap-up

Contact information



Dr. Christian A. Mahringer

TUM School of Management,
Campus Heilbronn

christian.mahringer@bwi.uni-stuttgart.de

Connect with me on [LinkedIn](#) for recent updates.

Generative AI was used to refine this presentation. All AI-generated content has been reviewed, and the author assumes full responsibility for its accuracy and integrity.